

Microscopy Core Gladstone Institutes Freezing Protocol

histology@gladstone.ucsf.edu

Fix tissue for 8-24 hours. Perfusion gives a much better result than drop-fixing. Use 20x the volume of fixative to tissue
Brains - Use longer fixation

Place in 10% sucrose for 1 hr

Place in 20% sucrose for 1 hr

Place in 30% sucrose O/N

Next Day

Prepare freezing equipment at least 20 minutes before you need it.
Place a beaker of isopentane (2-methyl butane) and surround with dry ice.
Place dry ice into the beaker until 1.5cm from top and slowly pour isopentane over the dry ice in the beaker. (see photo)
Wait until boiling stops and top off if necessary

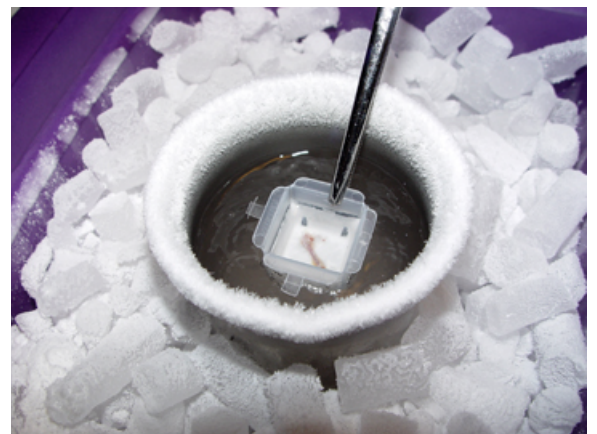
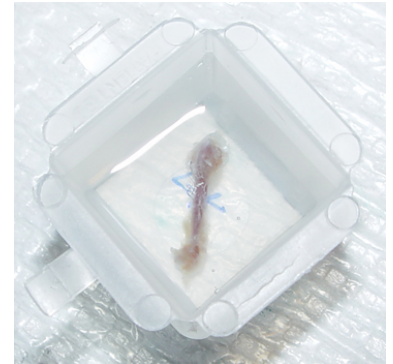
Fill an appropriately sized cryomold with a decent amount of OCT and orient tissue
(remember the initial cutting surface is the base of the mold)

Leave for 15mins . This helps prevent tissue curling away from the OCT in section when cut

Freeze tissue by floating OCT filled mold in the isopentane
Do not submerge!

Store samples in a labeled plastic bag at -80 until ready to section

Sign up for the cryostat
(if required)



This is the most common method of freezing. For muscle freezing contact core for more information

Sucrose gradient allows easier section cutting by cryoprotecting tissue against ice crystal formation